



MACHINE LEARNING · NLP · ACADEMIC PROJECT

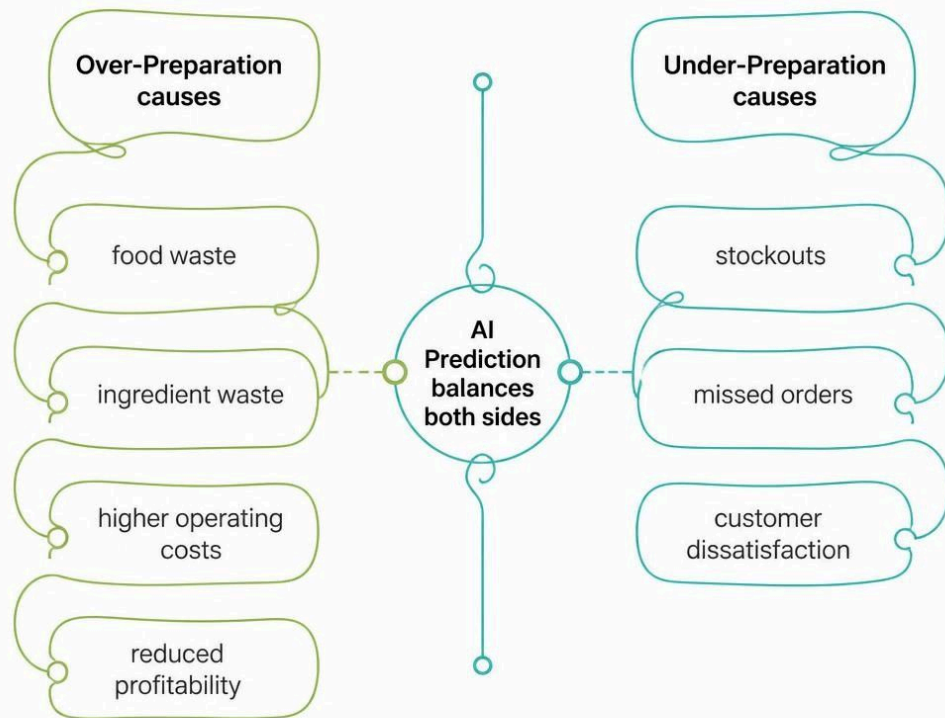
Restaurant Food Preparation Predictor

NLP + TF-IDF + XGBoost

Predict the right amount of food to prepare using dish descriptions, historical orders, and restaurant conditions.

Muhammed Ajnas · TRYCOD Tech School · Guide: Faisel / Fathima

The Core Problem



Project Objective

Build an **NLP + ML** system that predicts optimal dish preparation quantity.

- Dish name, description, category
- Day, month, season, weather
- Previous orders, ingredient stock
- Price, prep time, special events

❏ **Goal:** Improve preparation planning and reduce food waste.

Dataset & Feature Architecture

100,000 Restaurant Records

Target: Quantity_Prepared

NLP / Text

Dish_Name,
Description

Categorical

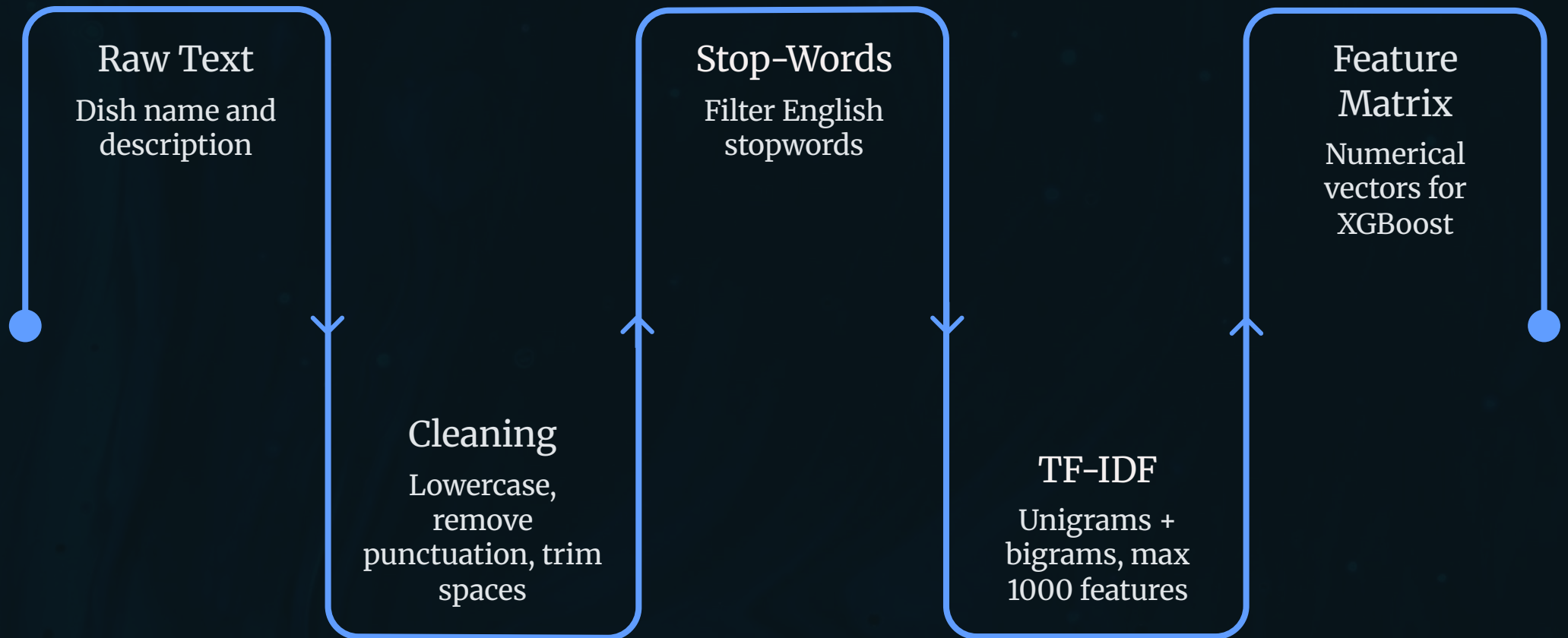
Category, Day,
Season, Weather,
Event

Numerical

Month, Price, Prep_Time, Previous_Orders, Stock

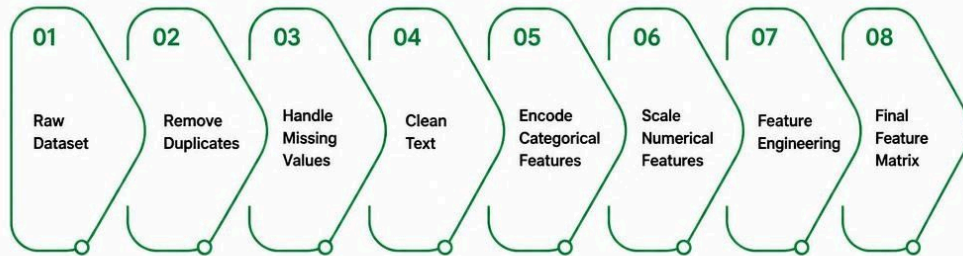
Dish Name	Description	Category	Previous_Orders	Qty Prepared
Chicken Biryani	Spiced rice with chicken	Main Course	142	138
Caesar Salad	Romaine, croutons, parmesan	Salad	67	65
Chocolate Lava Cake	Warm chocolate dessert	Dessert	89	92

NLP Pipeline: Text → ML Features



TF-IDF converts dish text into numerical features — e.g., *"Spicy chicken biryani with aromatic rice"* → ML-ready vector. Unigrams + bigrams capture both individual terms and meaningful phrases.

Data Preprocessing & Feature Engineering



Engineering Highlights

Log Transformation

`Previous_Orders` &
`Ingredients_Stock`

Weekend Indicator

`Day_of_Week` →
binary flag

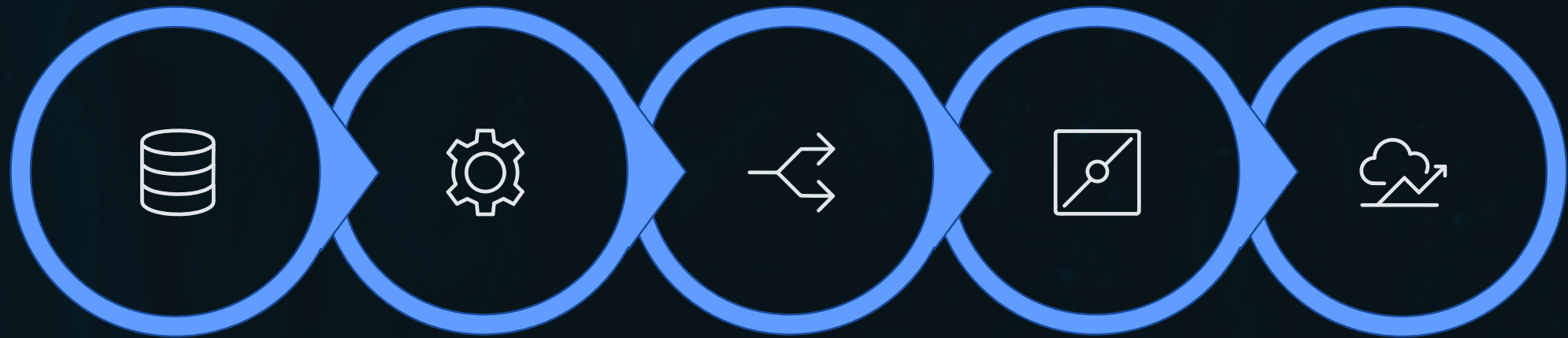
One-Hot Encoding

`Category`, `Season`,
`Weather`, `Event`

Standard Scaling

`Price`, `Prep_Time`,
`Month`

ML Methodology: XGBoost Regression



Dataset

Feature Eng

Train-Test

XGBoost

Predict

Why XGBoost?

- Handles nonlinear relationships
- Mixed feature types (text + numeric)
- Efficient on large datasets
- Captures feature interactions

Feature Fusion

TF-IDF Text + Categorical + Numerical + Engineered
→ XGBoost → Predicted Quantity

Model Evaluation & Validation

MAE

Mean Absolute Error

[FROM MODEL OUTPUT]

RMSE

Root Mean Squared Error

[FROM MODEL OUTPUT]

R^2 Score

Variance Explained

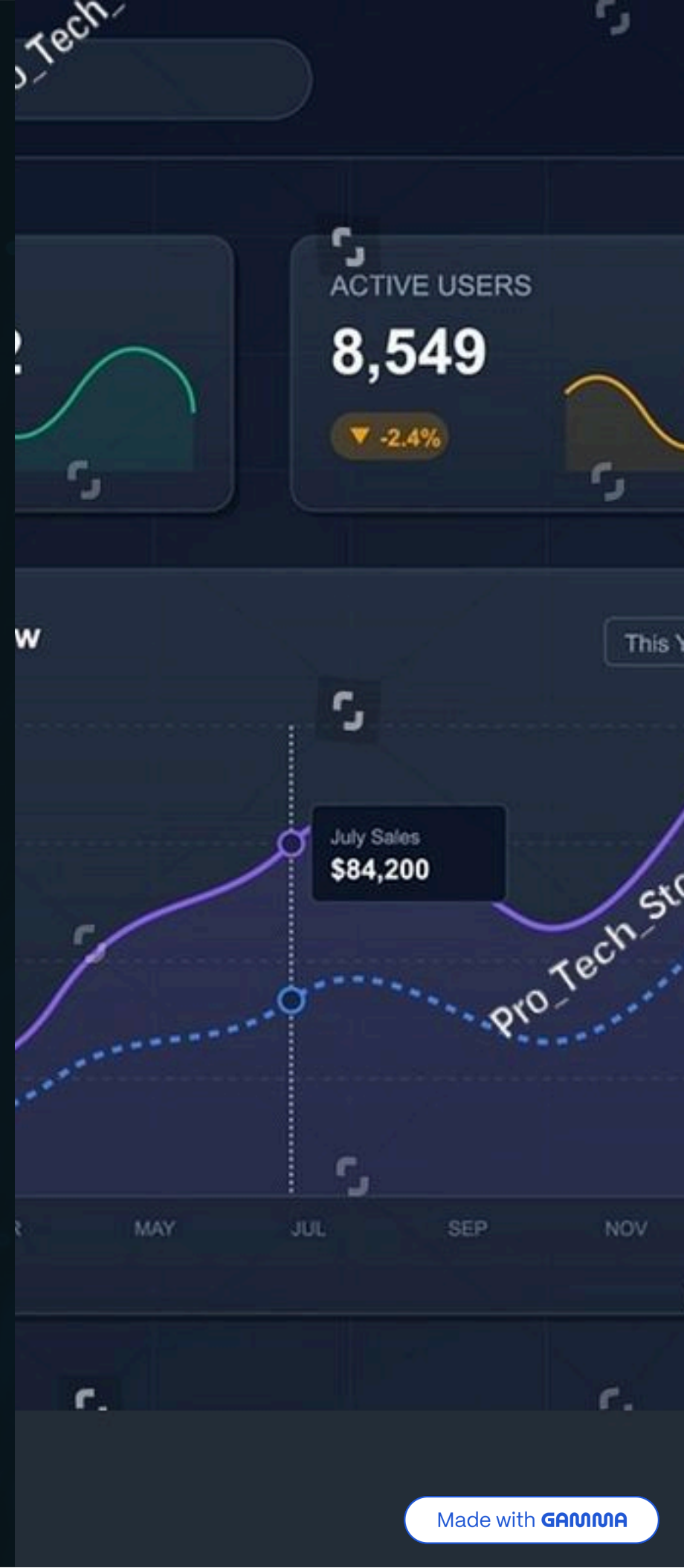
[FROM MODEL OUTPUT]

3-Fold CV

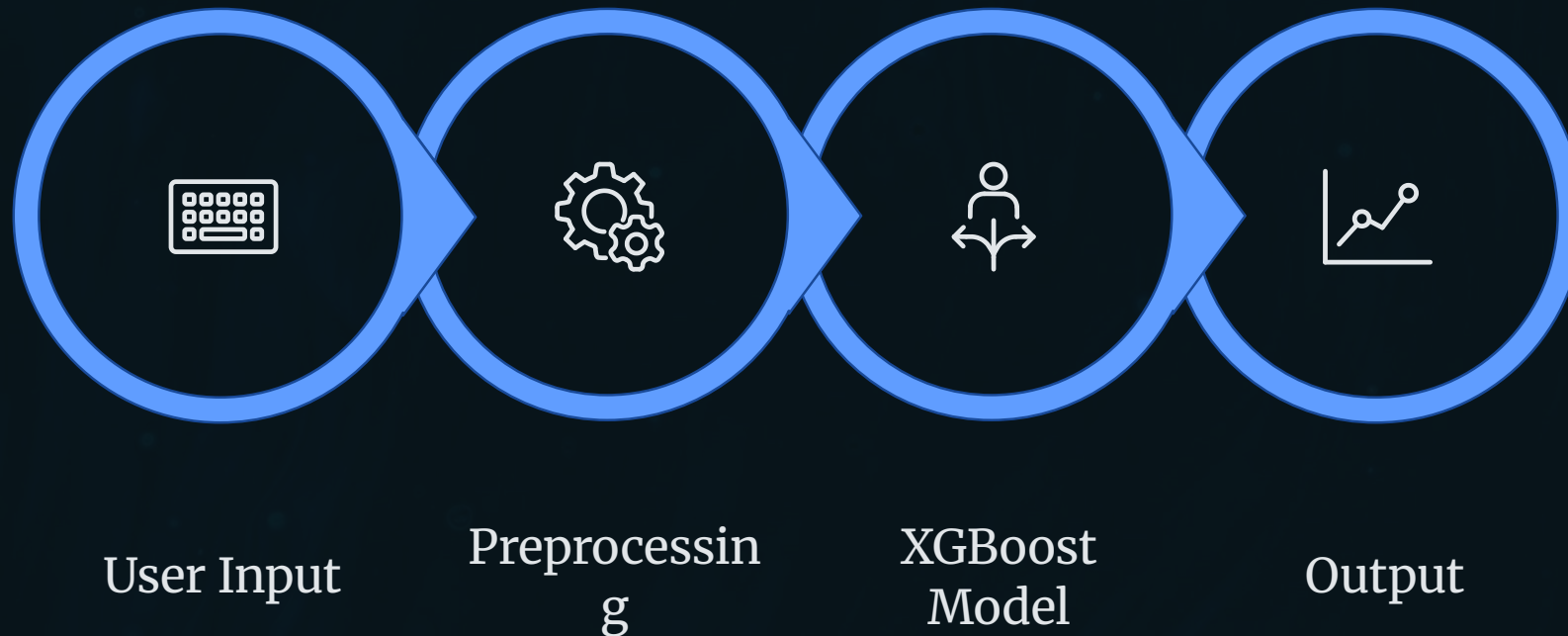
Average R^2 Score

[FROM MODEL OUTPUT]

Hyperparameter tuning: estimators · learning rate · max tree depth



The Prediction System



Interface: Gradio

Restaurant staff enter dish details and conditions → receive a preparation recommendation in seconds.

- ❏ **Example Output**
Recommended Preparation: **XX portions**
Expected Demand: **XX portions**

Real-World Impact & Future Scope

Who Benefits?



Owners & Managers

Cost control, inventory planning



Chefs

Informed production decisions



Customers

Better dish availability



Environment

Reduced food waste

Roadmap



Current System

TF-IDF + XGBoost



Advanced NLP

BERT, Sentence Transformers



Real-Time AI

Live orders, weather, waste tracking



Cloud & Mobile

Web app, mobile deployment

Conclusion

1

NLP / TF-IDF

Dish Name + Description

2

Feature Engineering

Restaurant & Historical Data

3

XGBoost Regression

Combined Features

4

Predicted Quantity

Recommended Preparation

A data-driven system that helps restaurants estimate **how much food to prepare** — supporting better planning and reducing unnecessary waste.

Future Direction: From prediction → to intelligent real-time restaurant decision support.

Thank You

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Questions & Answers